

Lab 3

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1.1:

Rain	Hagrid	Dumbledore	KB
F	F	F	F
F	F	T	F
F	T	F	F
F	T	T	F
T	F	F	F
T	F	T	T
T	T	F	F
T	T	T	F

Only one row makes all four facts true: rain = T, hagrid = F, dumbledore = T. Fact (4) forces dumbledore = T; fact (3) then forces hagrid = F (they are not both true); fact (1) says that if it is not raining then Hagrid is there, but Hagrid is not there, so it must be raining. `model_check(knowledge, rain)` checks every model in which the knowledge base is true and asks whether *rain* is true in all of them with dumbledore true. There is just one such model and rain is true in it, so the KB entails rain and the program prints True.

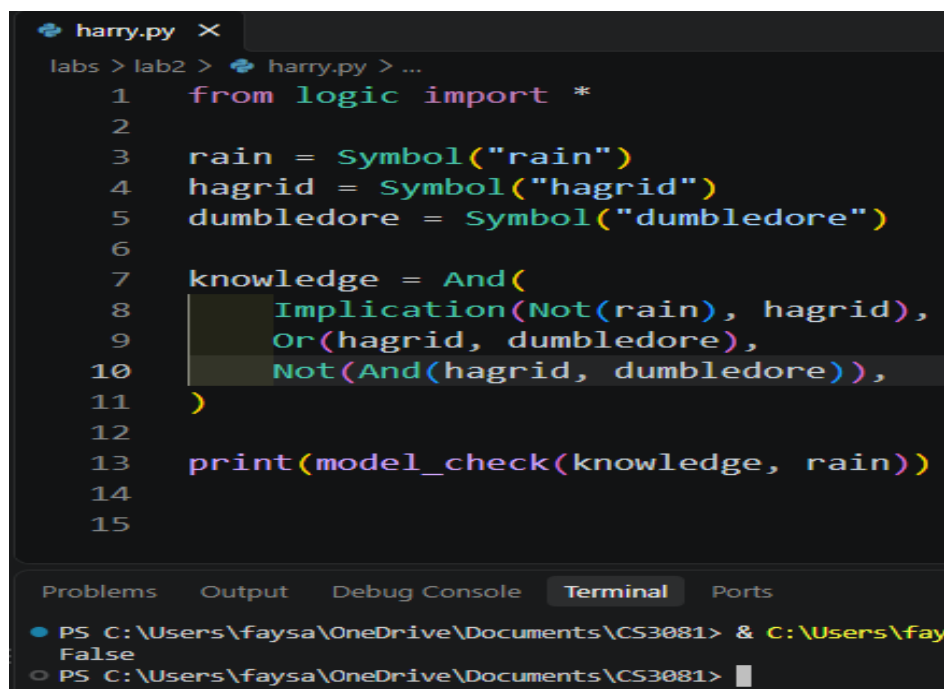
1.2:

Mustard and Plum are ruled out by known cards, so only Scarlet is left.

Kitchen and ballroom are ruled out, so only the library is left.

Revolver is ruled out. The unknown card says Scarlet, library and wrench cannot all be true; Scarlet and library are true, so wrench must be false. That leaves the knife

2.1:



```
harry.py X
labs > lab2 > harry.py > ...
1  from logic import *
2
3  rain = Symbol("rain")
4  hagrid = Symbol("hagrid")
5  dumbledore = Symbol("dumbledore")
6
7  knowledge = And(
8      Implication(Not(rain), hagrid),
9      Or(hagrid, dumbledore),
10     Not(And(hagrid, dumbledore)),
11 )
12
13 print(model_check(knowledge, rain))
14
15

Problems  Output  Debug Console  Terminal  Ports
PS C:\Users\faysa\OneDrive\Documents\CS3081> & C:\Users\fay
False
PS C:\Users\faysa\OneDrive\Documents\CS3081>
```

Without fact (4), several models stay possible, and rain will be true in some but not all of them, so the program should print False. From the truth table in Task 1, once the last column is removed three rows satisfy the first

three facts:

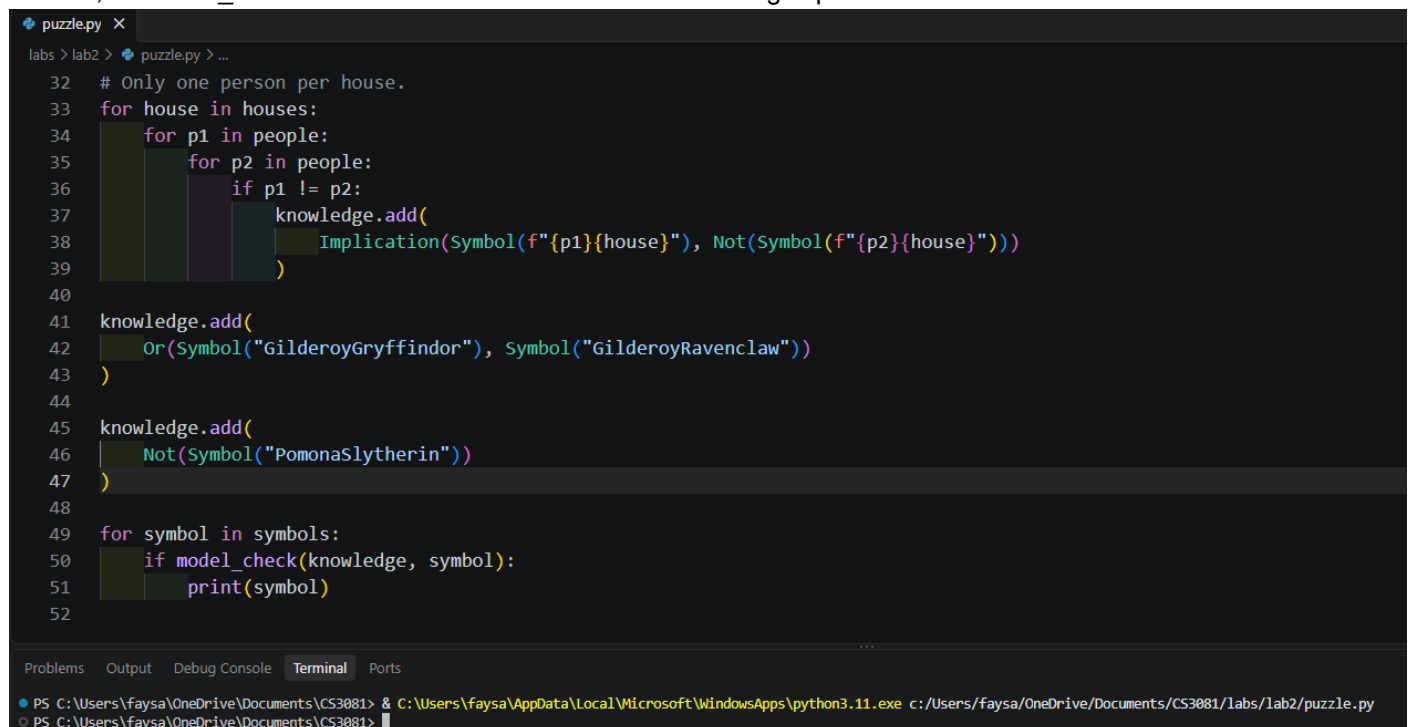
rain	hagrid	dumbledore	KB true?
T	T	F	TRUE
T	F	T	TRUE
F	T	F	TRUE

Rain is true in two of these models but false in the third (rain = F, hagrid = T, dumbledore = F: no rain, so Hagrid appears). We no longer care if harry is going to dumbledor, its now entailing rain to make the decision.

Knowledge Base now cares if harry goes hagrid or dumbledor when it rains but not both. But as default no rain should result harry going to hagrid.

2.2:

After deleting the clue, the remaining clues are: Gilderoy is in Gryffindor or Ravenclaw; Pomona is not in Slytherin. This leaves 8 valid assignments (Gilderoy G: Pomona H or R, then the other two people in either order; Gilderoy R: Pomona G or H, the other two in either order). Every symbol is false in at least one of these 8 models, so `model_check` returns False for all of them and nothing is printed.



```
puzzle.py X
labs > lab2 > puzzle.py > ...
32 # Only one person per house.
33 for house in houses:
34     for p1 in people:
35         for p2 in people:
36             if p1 != p2:
37                 knowledge.add(
38                     Implication(Symbol(f"{p1}{house}"), Not(Symbol(f"{p2}{house}"))))
39                 )
40
41 knowledge.add(
42     Or(Symbol("GilderoyGryffindor"), Symbol("GilderoyRavenclaw"))
43 )
44
45 knowledge.add(
46     Not(Symbol("PomonaSlytherin"))
47 )
48
49 for symbol in symbols:
50     if model_check(knowledge, symbol):
51         print(symbol)
52
```

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PS C:\Users\Faysa\OneDrive\Documents\CS3081> & C:\Users\Faysa\AppData\Local\Microsoft\WindowsApps\python3.11.exe c:/Users/faysa/OneDrive/Documents/CS3081/labs/lab2/puzzle.py

PS C:\Users\Faysa\OneDrive\Documents\CS3081>

The new clue we can add is `Symbol("HoraceGryffindor")` (Horace is in Gryffindor). This will make the answer unique again: Horace takes Gryffindor, so Gilderoy (Gryffindor or Ravenclaw) must be Ravenclaw. Pomona cannot be Slytherin, and Gryffindor and Ravenclaw are taken, so Pomona is Hufflepuff. Minerva gets the last house, Slytherin

```
puzzle.py X
labs > lab2 > puzzle.py > ...
34     for p1 in people:
35         for p2 in people:
36             if p1 != p2:
37                 knowledge.add(
38                     Implication(Symbol(f"{p1}{house}"), Not(Symbol(f"{p2}{house}"))))
39                 )
40
41     knowledge.add(
42         Or(Symbol("GilderoyGryffindor"), Symbol("GilderoyRavenclaw"))
43     )
44
45     knowledge.add(
46         Not(Symbol("PomonaSlytherin"))
47     )
48
49     knowledge.add(
50         Symbol("HoraceGryffindor")
51     )
52
53     for symbol in symbols:
54         if model_check(knowledge, symbol):
55             print(symbol)
```

Problems Output Debug Console Terminal Ports

PS C:\Users\faysa\OneDrive\Documents\CS3081> & C:\Users\faysa\AppData\Local\Microsoft\WindowsApps\python3.11.exe c:/Users/faysa/OneDrive/Documents/CS3081/labs/lab2/puzzle.py
GilderoyRavenclaw
PomonaHufflepuff
MinervaSlytherin
HoraceGryffindor
PS C:\Users\faysa\OneDrive\Documents\CS3081>

3.1:

```
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PS C:\Users\faysa\OneDrive\Documents\CS3081> & C:\Users\faysa\AppData\Local\Microsoft\WindowsApps\python3.11.exe c:/Users/faysa/OneDrive/Documents/CS3081/labs/lab2/knights.py
Puzzle 0
A is a Knight False
A is a Knave True
Puzzle 1
A is a Knight False
A is a Knave True
B is a Knight True
B is a Knave False
Puzzle 2
A is a Knight False
A is a Knave True
B is a Knight True
B is a Knave False
PS C:\Users\faysa\OneDrive\Documents\CS3081>
```

```

1  from logic import *
2
3  AKnight = Symbol("A is a Knight")
4  AKnave = Symbol("A is a Knave")
5  BKnight = Symbol("B is a Knight")
6  BKnave = Symbol("B is a Knave")
7
8  statement = And(AKnight, AKnave)
9  knowledge0 = And(
10     Or(AKnight, AKnave),
11     Not(And(AKnight, AKnave)),
12     Implication(AKnight, statement),
13     Implication(AKnave, Not(statement)),
14 )
15
16 print("Puzzle 0")
17 print(AKnight, model_check(knowledge0, AKnight))
18 print(AKnave, model_check(knowledge0, AKnave))
19
20 statement = And(AKnave, BKnave)
21 knowledge1 = And(
22     Or(AKnight, AKnave),
23     Not(And(AKnight, AKnave)),
24     Or(BKnight, BKnave),
25     Not(And(BKnight, BKnave)),
26     Implication(AKnight, statement),
27     Implication(AKnave, Not(statement)),
28 )
29
30 print("Puzzle 1")
31 print(AKnight, model_check(knowledge1, AKnight))
32 print(AKnave, model_check(knowledge1, AKnave))
33 print(BKnight, model_check(knowledge1, BKnight))
34 print(BKnave, model_check(knowledge1, BKnave))
35
36 same = Or(And(AKnight, BKnight), And(AKnave, BKnave))
37 different = Or(And(AKnight, BKnave), And(AKnave, BKnight))
38 knowledge2 = And(
39     Or(AKnight, AKnave),
40     Not(And(AKnight, AKnave)),
41     Or(BKnight, BKnave),
42     Not(And(BKnight, BKnave)),
43     Implication(AKnight, same),
44     Implication(AKnave, Not(same)),
45     Implication(BKnight, different),
46     Implication(BKnave, Not(different)),
47 )
48
49 print("Puzzle 2")
50 print(AKnight, model_check(knowledge2, AKnight))
51 print(AKnave, model_check(knowledge2, AKnave))
52 print(BKnight, model_check(knowledge2, BKnight))
53 print(BKnave, model_check(knowledge2, BKnave))

```

I used claude to generate the code

